

Diagnosis of devices & drivers

The first step to take for diagnosis is to know the list of hardware installed on your server. These details can be easily obtained using the command `ls`. This command goes further by- `lspci`, `lsblk` and `lscpu`. Let's us understand this by individual examples.

- The output of command `lsblk` is

```
# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
xvda        202:0    0   50G  0 disk
├─xvda1     202:1    0    1M  0 part
└─xvda2     202:2    0   50G  0 part /
xvdb        202:16   0   20G  0 disk
└─xvdb1     202:17   0   20G  0 part
```

In case, the command `ls` fails to expose the errors then you have to use the 'init' processes. This will enable you to inspect the Linux server function. The init process bootstraps the space. This also enables control over the processes running on a number of systems. In the below image you will see the result of `systemctl` command.

```
# systemctl status
* bastion.f347.internal
   State: running
   Jobs: 0 queued
  Failed: 0 units
 Since: Wed 2018-11-28 01:29:05 UTC; 2 days ago
  CGroup: /
          └─1 /usr/lib/systemd/systemd --switched-root --system --deserialize 21
              └─kubepods.slice
                  └─kubepods-pod3881728a_f2af_11e0_af77_06af52f87498.slice
                      └─docker-88b27385f4bae77bba834fdb60a61d19026bae13d10eb147783ae27819c
                          └─23860 /opt/bridge/bin/bridge --public-dir=/opt/bridge/static --c
                              └─docker-a4433f8d523c7e5bc772ec4db1061e4fa56c4e63a2d48f6bc831458c2ce
                                  └─23629 /usr/bin/pod
```

Following the logging

If you want to check the warning message or error in the kernel then use the command `Dmesg`. The output of this command is as shown in the first image on the next page.

If you want to see all the system logs in the message or log file, then you can monitor it with `tail` command. For this, you have to modify the hardware and perform some additional mounting. The output of this command i.e. `tail -f messages/log` is seen in the second image on the next page.